

⚡ SPIRAL × Hack-Nation Challenge 02

Agora

A Lightning-paid agent marketplace

8

BUILD PHASES

144 / 144

TESTS GREEN

~200 ms

PER PAID CALL

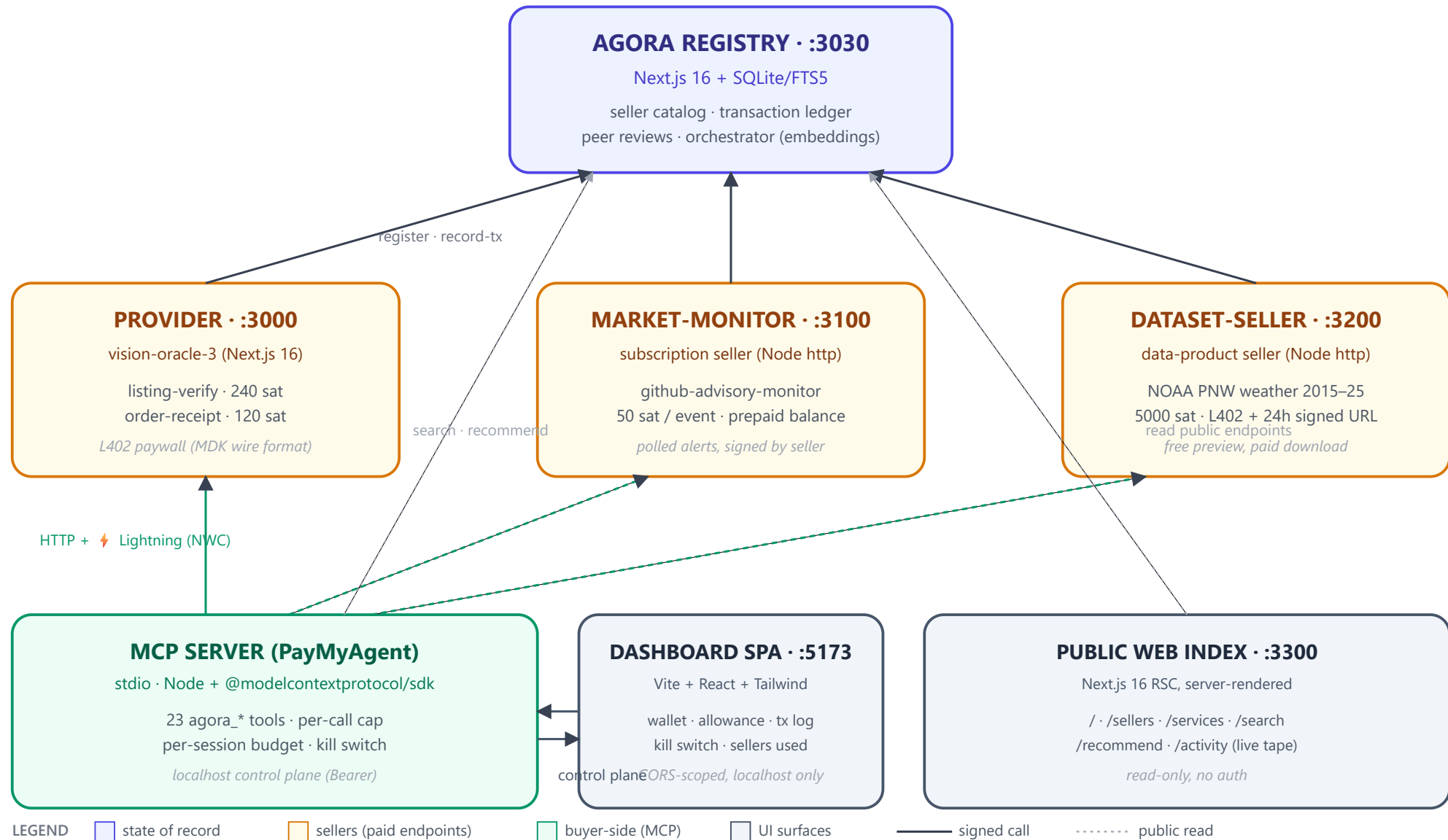
0

ACCOUNTS

github.com/Geoshua/agora · 2026-04-26

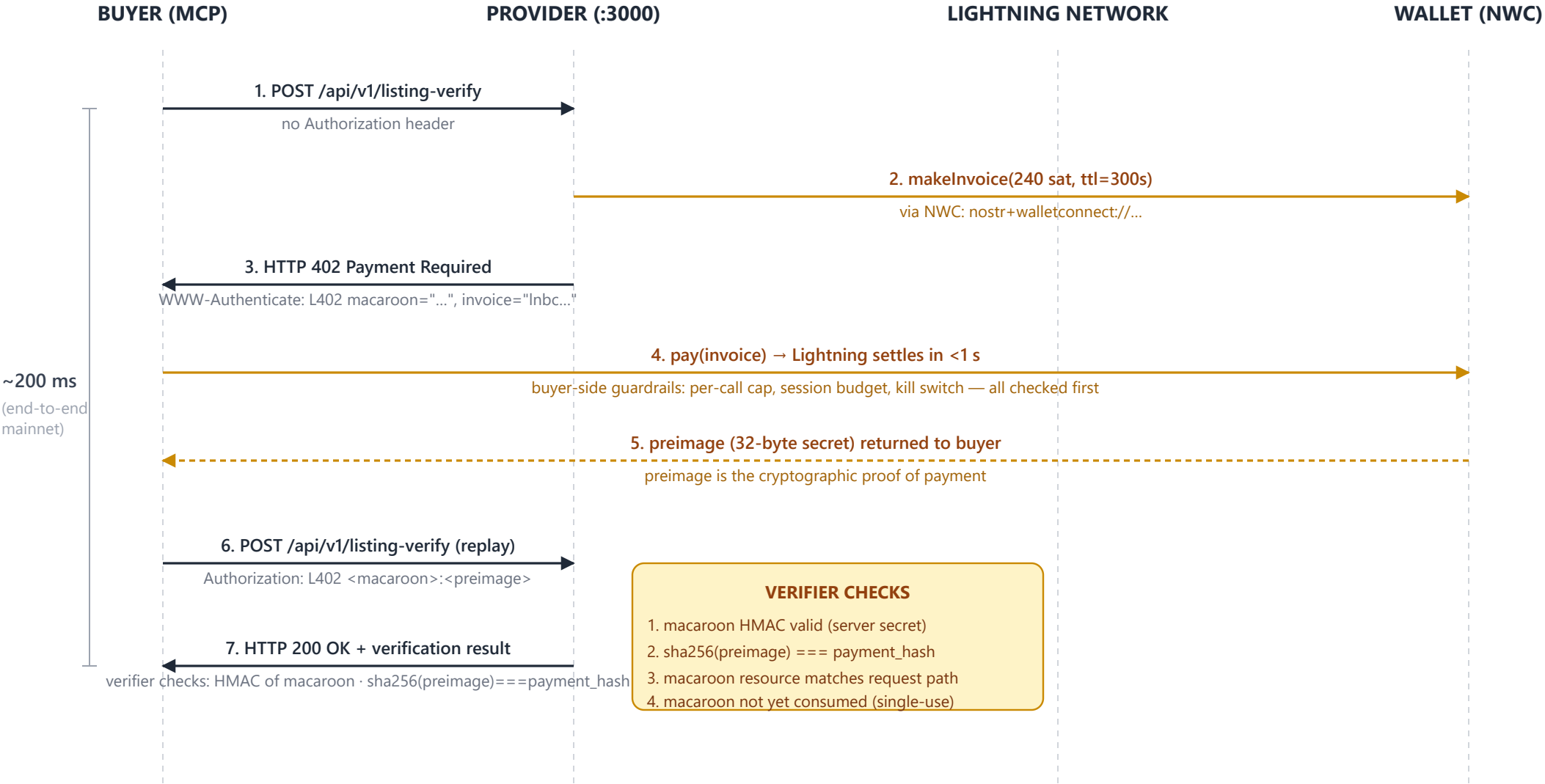
Agora — architecture overview

a Lightning-paid agent marketplace · 5 components, signed calls, no accounts



L402 round-trip · 402 → invoice → pay → replay → 200

~200 ms end-to-end on Lightning mainnet · no accounts, no API keys



In mock mode, steps 2/4/5 short-circuit through /api/dev/pay — same wire format, no sats move.

Macaroon wire format is byte-compatible with MoneyDevKit's mdk402-token-v1.

Identity = Ed25519 keypair · every call is signed

no email · no password · no OAuth · the public key IS the agent ID

SENDER (e.g. seller registering)

1. canonical string

```
POST
/api/v1/sellers/register
sha256(body) · unix-ms timestamp
```

2. sign with private key

```
sig = ed25519.sign(string, sk)
```

private key never leaves the agent

HTTP

3 headers carry identity + proof

X-Agora-Pubkey:

e72536984944304a6e615021c95f7207...

X-Agora-Timestamp:

1745683425181

X-Agora-Sig:

a1b2c3... (64-byte Ed25519 signature)

RECEIVER VERIFIES (Registry, Provider, Agent — anywhere a signed call lands)

1

Reconstruct canonical string

from request (METHOD, PATH, sha256(body), timestamp from header)

2

Verify signature · `ed25519.verify(sig, string, X-Agora-Pubkey)`

if false → 401. The pubkey IS the identity claim — no lookup, no DB.

3

Check timestamp window · `| now – timestamp | < 5 minutes`

prevents replay of captured signed calls outside the freshness window

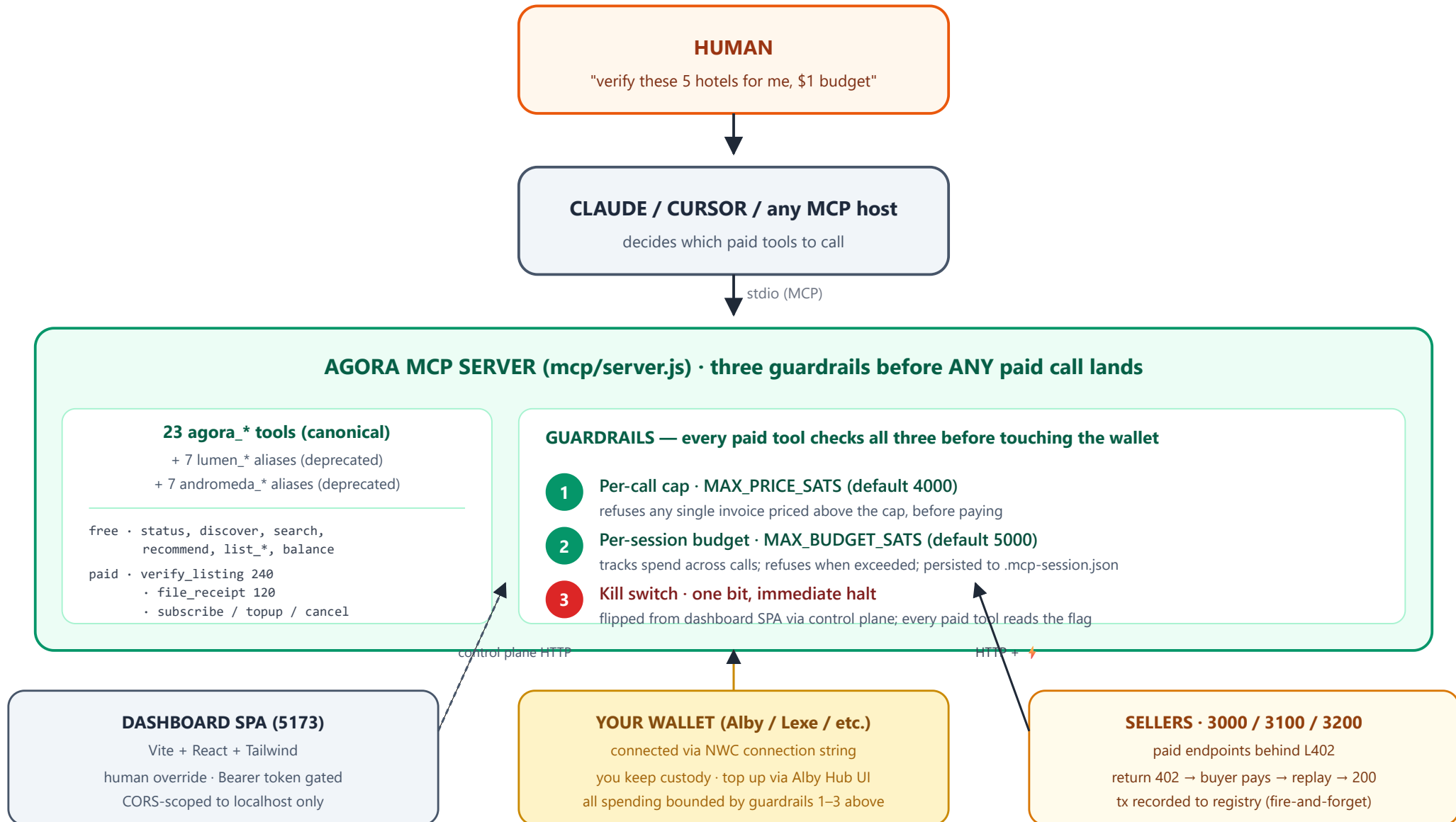
✓

Accept request as authentically from this pubkey

~~Registry stores pubkey as the seller's globally unique ID — no accounts created~~

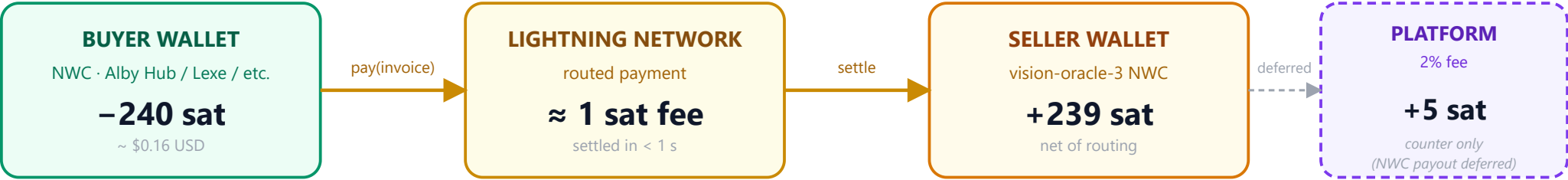
PayMyAgent — the human-to-agent bridge

Give Claude a sat budget and a goal. The MCP server enforces three guardrails between the agent and your wallet.



Money flow · one paid call · buyer → Lightning → seller (+ ledger)

Sub-cent amounts · ~200 ms settle · cryptographic proof of payment · no chargebacks



REGISTRY LEDGER (transactions table) · fire-and-forget POST after every settle

POST /api/v1/transactions/record · signed by seller

```
{ buyer_pubkey: "9f4a...b201"
  seller_pubkey: "e725...4af7" (← cryptographically the signer)
  service_id: "vision-oracle-3.listing-verify"
  amount_sats: 240, platform_fee_sats: 5, payment_hash: "..." }
```

→ /activity feed updates within 2 s · /sellers/<pubkey>/stats reflects new totals

Same wire format · two execution modes

MOCK MODE · default

- Deterministic preimages generated in-process
- POST /api/dev/pay returns the preimage instantly
- Counter-based "wallets" — no NWC client constructed
- Same L402 flow, same macaroon shape, same headers
- Zero network egress to payment infrastructure

Ideal for: CI, demos, offline development, SPIRAL judging

MOCK_MODE=true (default everywhere)

Failure modes: invoice expires (TTL 5 min) · preimage doesn't match (provider rejects) · macaroon already consumed (single-use enforced)

REAL MODE · Lightning mainnet

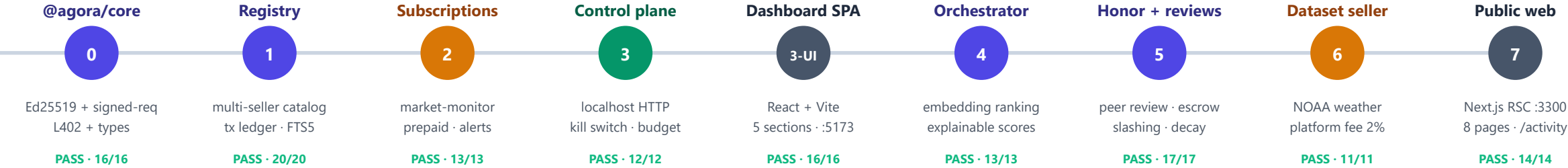
- Provider mints invoice via NWC (makeInvoice 240-1000 msat)
- Buyer pays via NWC (LNClient.pay, returns real preimage)
- Provider verifies sha256(preimage) === payment_hash
- Visible on Alby Hub / Lexe transaction logs both sides
- Settles in 200-800 ms · ~1 sat routing fee typical

Bring-your-own-wallet · NWC string in .env, that's the whole setup

MOCK_MODE=false + NWC_URL=nostr+walletconnect://...

Build progression · 8 phases · 144/144 tests green

each phase ships with a test gate · existing endpoints frozen · additive-only



⚡ Post-phase deltas

MDK migration (ADR 0014): macaroon wire format byte-aligned with mdk402-token-v1 · soft-transition verifier accepts legacy + new shapes

🚀 Deploy hardening

Fly.io configs for registry + web · Dockerfiles · admin-secret enforcement (default fallback removed) · DEPLOY.md runbook

📡 Live activity feed

GET /api/v1/transactions/recent (public, schema agora.transactions.v1) · /activity page on web (live tape, 2s cache)

PHASE COLOR · ROLE

● Registry / shared ● Sellers ● MCP / buyer-side ● UI surface

All gates green: 144/144 across 10 test scripts

60-second demo script

≈ 145 words · 5 beats · read off this page or the rendered slides

BEAT 1 — HOOK [01-architecture]

10 S

Agora is a marketplace where AI agents pay each other over the Lightning Network — no accounts, no API keys, no checkout. Five services: a registry, three sellers, a Claude bridge.

BEAT 2 — THE ATOM [02-l402-flow]

15 S

One paid call takes 200 milliseconds. The agent posts a request — gets back HTTP 402 and a Lightning invoice. Pays it. Replays the request with the preimage as proof. Server verifies, returns the result. That's the whole protocol — it's called L402.

BEAT 3 — NO ACCOUNTS [03-identity-signing]

10 S

Identity is an Ed25519 keypair. Every agent generates one on first run. The public key IS the agent's global ID. Every cross-service call is signed. There is no users table.

BEAT 4 — THE AGENT UX [06-mcp-bridge]

10 S

You give Claude a goal and a sat budget. A per-call cap, a per-session budget, and a kill switch sit between Claude and your wallet. You stay in custody.

BEAT 5 — CLOSE [04-phases]

15 S

Eight build phases, 144 of 144 tests green, three independent security audits. L402 wire format is byte-aligned with MoneyDevKit. Mock mode runs offline; flip two flags and you're on Lightning mainnet. **The repo is live. The marketplace works today.**

Numbers + commands

NUMBERS WORTH QUOTING

Per-call latency (mainnet)	~200 ms
Cheapest paid call (order receipt)	120 sat
Most expensive paid call (NOAA dataset)	5000 sat
Smallest unit cards can price	~30¢
Build phases shipped	8
Test gates green	144 / 144
MCP tools registered	30
Lightning fee on a 240-sat call	~1 sat

TRY IT AFTER THE DEMO

```
git clone git@github.com:Geoshua/agora.git cd agora npm run install:all
npm run demo:multi
```

Mock mode default — no wallet needed. ~200 ms × 2 paid calls.
For mainnet: paste two NWC strings + flip MOCK_MODE=false.

REPO

github.com/Geoshua/agora

8 ADRs · 6 diagrams · 3 audit reports · live activity feed at /activity